

IM3180 Design and Innovation Project (AY2025/26 Semester 1)

Individual Report

Name: Quek Zhi Yu

Group: IE08

Project Title: DeciWise

Contributions to the Project and Report (1-2 pages)

Project

- Initial project setup (Git, React)
- Oversaw the technical side of the entire project, coordinating teammates' progress, managing updates
- In charge of git merges, ensuring things work well together and also debugging for errors
- Drafting of Initial Pages (Admin, Login , Dashboards)
- Flowchart Editor (React Flow integration, node creations and interactions, layout algorithm, Flow Debugger)
- EditorToolSideBar (Drag-and-drop of new nodes, AI generated suggestion node options-fetching api to generate suggestion)
- Scene Editor (Passing flowchart data to backend presenting data in nodes to regenerate nodes, temp image workflow to store large b64 images from gemini)
- Implemented local caching using localforage, and resolved multiple UI/UX issues related to layout, rendering, and state management.
- Integrate backend to frontend (Mainly ScenarioPrompt, Flowchart Editor and Scene Editor)

Report (indicate pages and appendixes written by you for the Group Report)

- 3.4 Existing diagramming or flow visualization technologies
- 4.1 Design Consideration / Choice of Components / Tech Stack (Frontend)
- 4.3.1 Scenario Prompt
- 4.3.3 Flowchart Editor
- 4.3.4 Save & Load Functionality
- 4.3.5 Caching & Storage Strategy
- 4.3.8 Frontend–Backend Integration
- 4.4.1. Functional testing (node creation, editing, saving)
- 4.4.3. UI testing (drag/drop stability, text editing)
- 5.3. Handling missing links or repeated nodes
- 5.4. Managing React Flow state
- 5.5 Large image payloads crashing the DB

Other Deliverables like Project Charter (indicate pages and appendixes), Poster, Presentation Slides (indicate slide numbers) and Video, if applicable

Project Charter - Project Description

Reflection on Learning Outcome Attainment

Reflect on your experience during your project and the achievements you have relating to at least two of the points below:

- (a) Engineering knowledge
- (b) Problem Analysis
- (c) Investigation
- (d) Design/development of Solutions
- (e) Modern Tool Usage
- (f) The Engineer and the World
- (g) Ethics
- (h) Individual and Collaborative Team Work
- (i) Communication
- (j) Project Management and Finance
- (k) Lifelong Learning

Point 1: Modern Tool Usage

During this project, I have spent most of my time working in a full stack environment using React, Flask, MongoDB and AI Api keys. Initially, switching between backend and frontend was very confusing, having to call APIs and making sure that data returns properly on the frontend. It took me a lot of trial and error before getting the hang of it and really understanding how the data moves through a full-stack system. Now I can say that I know how to connect the frontend component to the backend, how to fetch from the database and also how to debug when there are errors. I also got the chance to learn how to use AI APIs in this project, it was a brand new experience for me seeing how my teammates work the AI portion code. I had imagined it to be way harder but surprisingly it was quite comprehensible. Since I was the one mainly integrating the AI returns to the frontend I also had the chance to fix the prompts and see how the changes came along. It was pretty amazing. Learning how to integrate AI into our system frontend was one of the great takeaways I had.

For the frontend side, I also had to learn how to make use of third-party libraries like React Flow, Dagre, Framer Motion etc. These were really crucial to our project as our entire frontend does rely on a good flowchart editor which was why I spent weeks on perfecting the generation of the nodes and edges, ensuring that they were user friendly and fully customisable. It was perfect that React Flow and Dagre existed as it really made my life a lot easier even though I still had to go in depth to learn how the different layers worked. I felt that I have improved quite a bit in debugging through making use of console logs and browser inspection tools and network tab. It was hard to decipher what was going wrong at the start but slowly as I debug more types of errors, I was getting familiar with the different types of errors that were showing and I would know where to go and fix..

I also had a better understanding of version control through Git and Github from this project.

Previously in IDIP, it was version control between 2 people but this time round we had 5 people working on development at the same time. Despite having some prior experience on the branching and merging, there were still errors, especially when most of us are not experienced in it. That resulted in many nights of debugging to ensure that everyone's work was not lost. The idea of each person having a branch each was not bad as it was clear of who had worked on what, however I did find the flaw when we started merging each other branch into our own, it overwrites our content. I have learnt that individual branching might not be the best method and instead we should have tried doing feature branching where whenever each of us work on a new feature, we will create a new branch. This would have ensured better version control amongst so many members. Another thing

was having a development branch. I personally had made the mistake of pulling people's branch into my own and I ended up having to fix all the integration problems. This was quite inefficient as everyone was held up by me to fix the branch before they could continue. Hence some main take away from Git branching was that we should always commit small changes so that it's easier to track as well as pushing and pulling from the development branch to ensure less errors.

Overall this project taught me a lot about how important are modern tools in software development. With all the libraries, complicated features become possible as long as you know how to use the tools properly. I have realised that software development was not just about coding but also understanding what are the available tools and integrating them effectively to achieve our goals.

Point 2: Individual and Collaborative Team Work

Through this project I have discovered the way to work independently while also working with my team. For the individual parts, I took charge of the major parts of the system which were the flowchart and scene editors, the back to frontend integration, as well as the Git merging workflow. I had to take initiative to figure out not only problems of my own as well as problems that my teammates were facing. I had to quickly pick up new things that were added, things that weren't working, flaws in the system and update my team accordingly.

On the team work side, I constantly communicated with my teammates about technical changes, helped them with Git when they got stuck, and resolved merge conflicts so that everyone's efforts can work properly together. I would help to explain how certain parts of the system work or help teammates debug their codes so that the project can continue smoothly. What I felt that I enjoyed the most was that we would be sharing knowledge and learning from each other. By reading their codes I was able to learn parts of the system that was not within my workscope. Through our discussions we were also able to come together to conclusions with one another's input to get to our goal.

I felt my team was really amazing as we had a great distribution of workload, everyone had their own set of contributions. Those working on the technical side had faced many challenges on our own but we would still cover for each other and help each other debug wherever needed. I love hearing "I fixed it!" Those that didn't work as much on technical aspects focused on areas like research, presentation, video. This really showed me how an efficient team worked well together when work is distributed well, everyone is proactive and supports one other. We could trust each other to improve the work to reach our goals as a team.

Overall this project really taught me how to balance being independent and playing my part as a team player. I learnt that I can be coding, fixing problems, or even coordinating and initiating to make sure everyone works come together in the end.

Acknowledging/Declaring the Use of GAI

Please refer to NTU's Current Policy & Guidelines on the Use of Generative AI available in NTUlearn home page and the link:

[https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-\(GAI\)-Tools-on-Education.aspx](https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-(GAI)-Tools-on-Education.aspx)

- Complete the following declaration if applicable.
- Create a Paper Trail to document the input prompt, output obtained, and how you have used it

I Quek Zhi Yu (student name),
quek0248@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission:

1. Name of course: Design & Innovation Project
2. Course Code: IM3180
3. Instructor: Prof Andy Khong
4. Title of Assignment/Project Submission: DIP Individual Report

In relation to the foregoing I hereby declare that, fully and properly in accordance with the Assignment/Project Instructions I have (check where appropriate):

- i. Used GAI as permitted to assist in generating key ideas only. ☐
- ii. Used GAI as permitted to assist in generating a first text only. ☐
- And/or
- iii. Used GAI to refine syntax and grammar for correct language submission only. ☐
- Or
- iv. As it is not permitted: Not used GAI assistance in any way in the development or generation of this assignment or project. ☒

I also declare that I have :

- a. Fully and honestly submitted the digital paper trail required under the assignment/project instructions; and that
- b. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote; and that,
- c. Apart from the foregoing notices, the submission is wholly my own work.

Quek Zhi Yu 
Student Name & Signature

13/11/2025
Date